

## **Project Title** 3D Deep Sea Rescue: Submarine Navigator

**Project Description** This project is a fully 3D underwater exploration game developed using OpenGL and GLUT. The player pilots a deep-sea submarine through a complex cavern system to locate and collect stranded rescue pods before their oxygen supply runs out.

Unlike standard 2D-plane driving games, this project focuses on full volumetric movement, environmental physics, dynamic visual effects, and hazard management. The player must balance spatial awareness, resource management, and precise maneuvering to survive the deep ocean.

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## **Core Features (15 Total)**

### **Group 1: Environment & Atmosphere**

1. **3D Cavern Network:** A procedural-style or manually mapped underwater maze built using OpenGL primitives (scaled cubes for walls, cylinders for pillars).
2. **Depth-Based Fog (GL\_FOG):** The environment dynamically becomes darker and foggier as the submarine descends on the Z-axis, simulating deep-sea pressure and darkness.
3. **Procedural Bubble Particles:** Arrays of tiny spheres that constantly spawn and drift upwards, providing a realistic underwater visual aesthetic.
4. **Submarine Headlights (GL\_LIGHTING):** A directional spotlight pointing forward from the submarine to cut through the fog and illuminate the path ahead.

**Group 2: Physics & Movement** 5. **6-Degree-of-Freedom (6-DOF) Hydrodynamics:** Multi-axis movement (pitch, yaw, elevation, and thrust). Incorporates fluid friction so the submarine glides smoothly to a halt rather than stopping instantly. 6. **Dynamic Ocean Currents:** Invisible spatial zones in the map that act as "wind tunnels." Entering these zones pushes the submarine off course, requiring the player to counter-steer. 7. **Dual Camera System (gluLookAt):** A toggleable camera system allowing the player to switch between a cinematic 3rd-person "chase" view and a tense 1st-person cockpit view.

**Group 3: Objectives & Survival** 8. **The Oxygen Timer:** A constantly depleting Oxygen meter that acts as the game's countdown timer and core fail-state condition. 9. **Rescue Pods (Primary Goal):** Stranded pods scattered throughout the 3D maze. Collecting these increases the score and is required to clear the level. 10. **Bioluminescent Sponges (Survival Pickups):** Optional glowing items placed in risky areas. Collecting them instantly restores a chunk of the Oxygen meter. 11. **Upgrade Crates:** Rare collectibles that permanently upgrade the submarine's stats (e.g., faster turning speed or slower oxygen depletion rate).

**Group 4: Dangers & Combat** 12. **3D Bounding Collisions:** Full 3D bounding-box/sphere collision detection against the cavern walls, floor, and ceiling. Crashing applies a heavy oxygen

penalty. 13. **Patrolling Hazards:** Moving obstacles (such as spiked sea mines or anglerfish) with basic tracking AI that slowly drift toward the player if they get too close. 14. **Destructible Terrain:** Fragile rock walls blocking certain paths. The player can fire torpedoes to blow them up and create shortcuts. 15. **Sonar Pulse Radar:** Since visibility is low, the player can trigger an expanding, transparent wireframe sphere. If the pulse hits a hidden item or enemy, it visually highlights it on the screen.

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## Unique Feature

**Hydrodynamic Movement & Environmental Physics Engine** Unlike standard driving simulators that only track X and Y coordinates with simple rotation, this system evaluates true 3D flight in a fluid environment. The submarine experiences "water resistance" and momentum. Combined with the **Dynamic Ocean Currents** and **Depth Fog**, the player must master balancing vertical buoyancy, drifting momentum, and low visibility simultaneously to navigate safely.

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## Individual Contributions

To ensure a balanced workload, the 15 features and core systems have been divided equally among the three team members:

### Member 1: Core Physics & Vision (The "Engine")

- **Movement:** Program the 6-DOF Hydrodynamic movement (pitch, yaw, thrust, fluid friction).
- **Physics:** Implement the Dynamic Ocean Currents logic to push the player.
- **Camera:** Write the dynamic `gluLookAt` logic for the Dual Camera System (1st/3rd person).
- **Lighting & Fog:** Set up the Submarine Headlights (`GL_LIGHTING`) and the Depth-Based Fog (`GL_FOG`).

### Member 2: Interactions & Hazards (The "Action")

- **Collisions:** Handle all 3D bounding-box math for walls, ceiling, and floor, including crash penalties.
- **AI:** Program the tracking logic for the Patrolling Hazards.
- **Combat:** Manage the lists for Torpedoes and Destructible Terrain, handling deletion upon impact.
- **Visual FX:** Code the expanding Sonar Pulse Radar and the Procedural Bubble particle system.

### Member 3: World Building & Game Loop (The "State")

- **Environment:** Design and build the actual 3D Cavern Network layout using primitives.
  - **UI & Rules:** Manage the Oxygen Timer, Game Over/Win states, and draw the 2D UI text.
  - **Objectives:** Place the Rescue Pods and write the logic to update the score/win condition.
  - **Pickups:** Place the Bioluminescent Sponges and Upgrade Crates, and write the logic to apply their effects when collected.
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### Tools & Technologies

- Language: Python
- Graphics Library: OpenGL (PyOpenGL)
- Utility Toolkit: GLUT